

	<u>magnetic field to the motion of the loop.</u>	
8(a)	Solar power collected = Intensity $\times$ Area $= (0.90 \times 10^3) \times 7.7$ $= 6930 \text{ W}$ Time of journey = $\frac{\text{distance}}{\text{speed}} = \frac{200}{67} \times 60 \times 60 = 1.07 \times 10^4 \text{ s}$ Total energy collected = power $\times$ time $= 6930 \times 1.07 \times 10^4$ $= 7.42 \times 10^7 \text{ J}$	[1] for power [1] for time [1] for ans
8(b)	Electrical energy converted = 26% of $7.45 \times 10^7 \text{ J}$ $= 0.26 \times 7.45 \times 10^7$ $= 1.9 \times 10^7 \text{ J}$	[1] correct %age [1] ans
8(c)(i)	For motor with 92% efficiency, $= \text{power required} = \frac{100}{92}(1100) = 1195 \text{ W} = 1200 \text{ W}$ Therefore, 1200 J of energy were being consumed each second.	[1] substitution [1] ans
8(c)(ii)	Discharge current $I = \frac{P}{V} = \frac{1200}{50} = 24 \text{ A}$	[1] substitution [1] ans
8(d)(i)	Number of solar cells is $N = \frac{7.7}{6.4 \times 10^{-3}} = 1.2 \times 10^3$	[1] ans
8(d)(ii)	Each cell collects an amount of energy equal to $\frac{7.45 \times 10^7}{1.2 \times 10^3} = 6.2 \times 10^4 \text{ J}$	[1] substitution [1] ans

8(e)(i)	Let the driving force and the resistive force be F and R respectively. $F = \frac{P}{v} = \frac{1100}{\left(\frac{67000}{3600}\right)} = 59 \text{ N}$	[1] correct formula and substitution [1] ans
8(e)(ii)	Since the speed is constant, net force is zero and thus the driving force is equal to the resistive force. $R = F \Rightarrow R = 59 \text{ N}$	[1] equating driving force to resistive force
8(f)(i), (ii)	See graph below.	
8(f) (iii)	The resistive force R is proportional to v^2 .	[1]
8(g)	Possible factors are (1) proximity of sun to earth (2) angle of elevation of sun's rays (3) degree of cloudiness (4) degree of trees cover along the route	[2] any two factors

8(f)(i), (ii) [1] for plotting the last point correctly.

[1] For drawing correct best-fit line.

